

PRINCIPLES OF FLIGHT

By : 1st Lt. Omar Adel El-Sayed

SECTION ONE

Inside an Aircraft (Onboard Systems)

Flight Control & Automation

- **Fly-by-Wire (FBW) Systems** – Electronic control of aircraft surfaces (no mechanical linkages).
- **Autopilot Systems** – Automated control for altitude, speed, and heading.
- **Automatic Flight Control System (AFCS)** – Stabilization and control for aircraft.

Avionics & Instrumentation

- **Flight Management System (FMS)** – Calculates optimal flight paths, fuel efficiency, and navigation.
- **Cockpit Display Systems (Glass Cockpit)** – Screens that replace analog instruments (PFD, MFD, EICAS).
- **Head-Up Displays (HUD) & Augmented Reality (AR) Displays** – Project flight data onto a transparent screen.
- **Engine Monitoring & Control Software** – Monitors fuel flow, thrust, temperature, and performance.
- **Air Data Computers (ADC)** – Processes airspeed, altitude, and pressure data.

Communication & Safety Systems

- **Aircraft Communication Addressing and Reporting System (ACARS)** – Digital communication between aircraft and ground.
- **Traffic Collision Avoidance System (TCAS)** – Alerts pilots of potential mid-air collisions.
- **Enhanced Ground Proximity Warning System (EGPWS)** – Prevents controlled flight into terrain.
- **Weather Radar & Lightning Detection Systems** – Predicts turbulence and storms.

Cabin & Passenger Systems

- **In-Flight Entertainment (IFE) Systems** – Screens, Wi-Fi, and media streaming for passengers.
- **Cabin Pressure & Climate Control Software** – Regulates cabin environment.
- **Aircraft Health Monitoring System (AHMS)** – Predictive maintenance based on sensor data.

Training & Flight Simulators

Full Flight Simulators (FFS)

- **Flight Dynamics Models** – Realistic physics for aircraft behavior.
- **Control System Emulation** – Simulates aircraft control responses.
- **Visual & Motion Systems** – High-resolution terrain and cockpit displays.
- **AI-Powered Virtual Instructors** – Adaptive training based on pilot behavior.

Part-Task Trainers (PTT)

- Simulated cockpit instruments for practicing specific tasks like **autopilot use or engine failure scenarios**

Augmented Reality (AR) & Virtual Reality (VR) Training

- Used for pilot training, **aircraft maintenance training**, and emergency procedures.

Air Monitoring & Navigation Systems

Air Traffic Management (ATM)

- **Air Traffic Control (ATC) Automation** – Helps controllers manage aircraft separation.
- **Automatic Dependent Surveillance-Broadcast (ADS-B)** – Real-time aircraft tracking.
- **Flight Data Processing Systems (FDPS)** – Manages aircraft routes, delays, and weather adjustments.

Navigation & Positioning

- **Global Navigation Satellite System (GNSS) & GPS** – Provides real-time location data.
- **Inertial Navigation Systems (INS)** – Uses gyroscopes and accelerometers for position tracking.
- **Flight Planning Software** – Calculates optimal routes based on fuel, weather, and airspace restrictions.

Weather & Environmental Monitoring

- **Numerical Weather Prediction (NWP) Models** – AI-based weather forecasting for flights.
- **Remote Sensing & LIDAR for Atmospheric Monitoring** – Detects turbulence and pollution.

Unmanned Aerial Systems (UAS) & Drones

Autonomous Flight & Control

- **Autopilot & Stability Augmentation Systems** – Keeps UAVs stable in various conditions.
- **Collision Avoidance Algorithms** – AI-driven obstacle detection and avoidance.
- **Swarm Intelligence Algorithms** – Coordination of multiple drones for military or commercial applications.

Ground Control & Mission Planning

- **UAV Ground Control Stations (GCS)** – Allows remote piloting and mission execution.
- **AI-Driven Image Recognition & Targeting** – Used in military, agriculture, and rescue operations.
- **Payload & Sensor Integration Software** – Controls cameras, LIDAR, and thermal imaging for reconnaissance.

Cybersecurity & AI Defense Systems

- **Drone Hacking & Anti-Drone Countermeasures** – Protecting UAVs from cyber threats.
- **Encryption & Secure Data Links** – Ensuring secure communication between drones and operators.

01

GENERAL DEFINITIONS



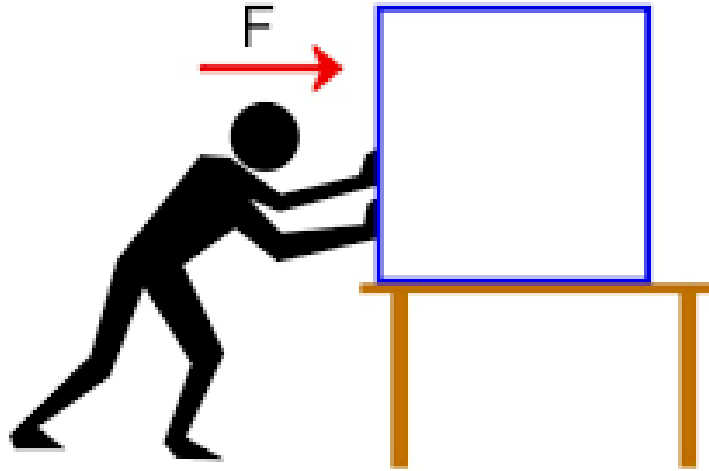
GENERAL DEFINITIONS

GENERAL DEFINITIONS

WHAT IS FORCE ??

GENERAL DEFINITIONS

FORCE : A push or pull on an object, causing it to move or change motion. Measured in newtons



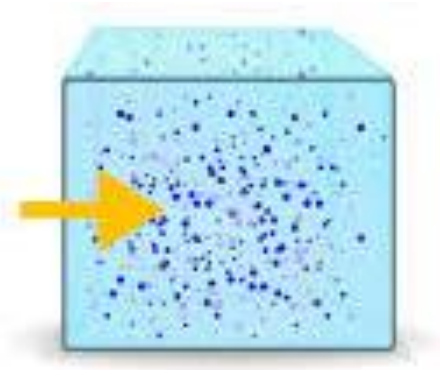
GENERAL DEFINITIONS

GENERAL DEFINITIONS

WHAT IS MASS ??

GENERAL DEFINITIONS

MASS : The amount of matter in an object, measured in kilograms.



GENERAL DEFINITIONS

GENERAL DEFINITIONS

IS MASS A FORCE ??

GENERAL DEFINITIONS

GENERAL DEFINITIONS

MASS is NOT a Force, Weight is a Force.

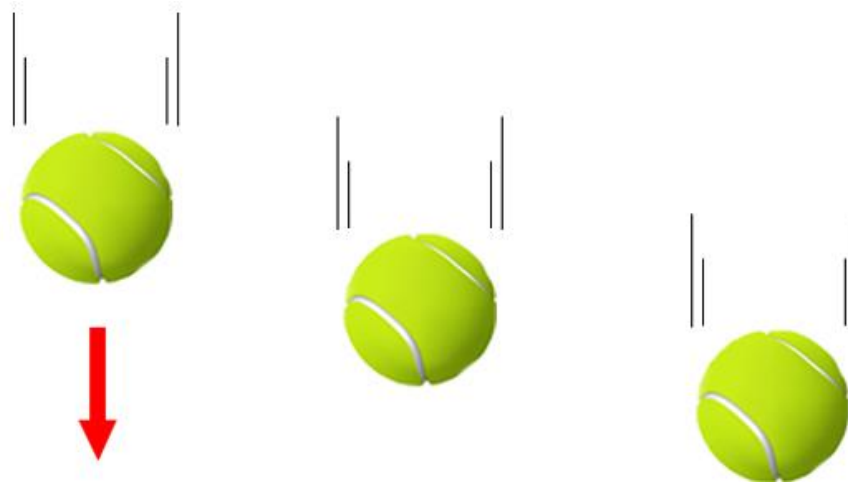
GENERAL DEFINITIONS

GENERAL DEFINITIONS

WHAT IS GRAVITY ??

GENERAL DEFINITIONS

GRAVITY : The force that attracts objects toward Earth, causing objects to fall or have weight.



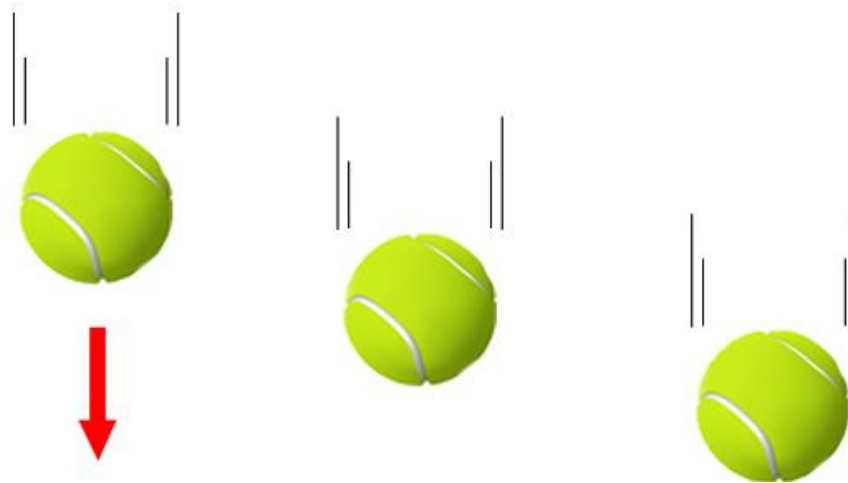
GENERAL DEFINITIONS

GENERAL DEFINITIONS

THEN WHAT IS WEIGHT ??

GENERAL DEFINITIONS

WEIGHT : The force exerted by gravity on an aircraft's mass (Weight = Mass x Gravity). It pulls the plane downward.



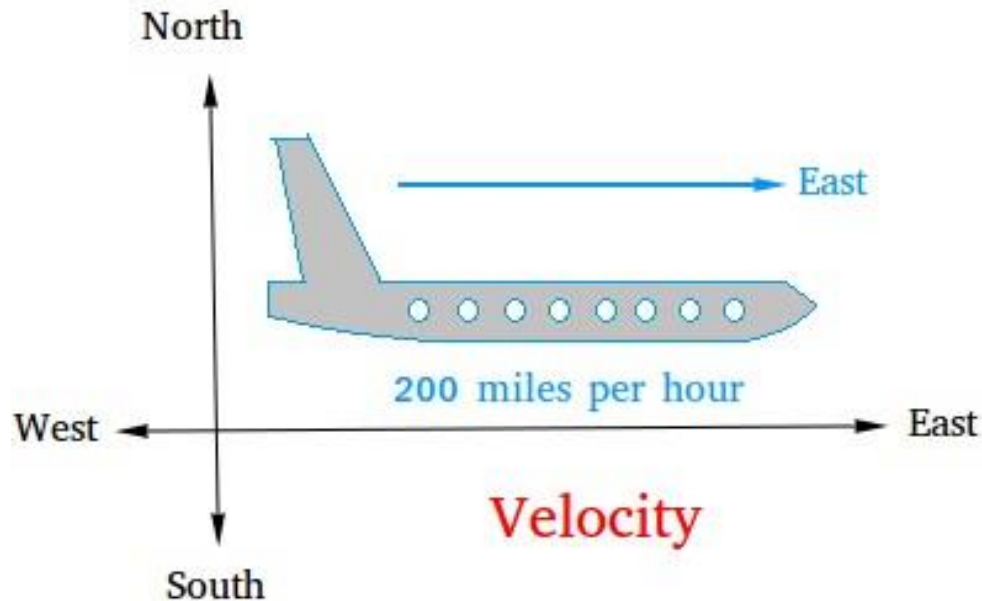
GENERAL DEFINITIONS

GENERAL DEFINITIONS

WHAT IS VELOCITY ??

GENERAL DEFINITIONS

VELOCITY : the rate of change of position in a certain direction



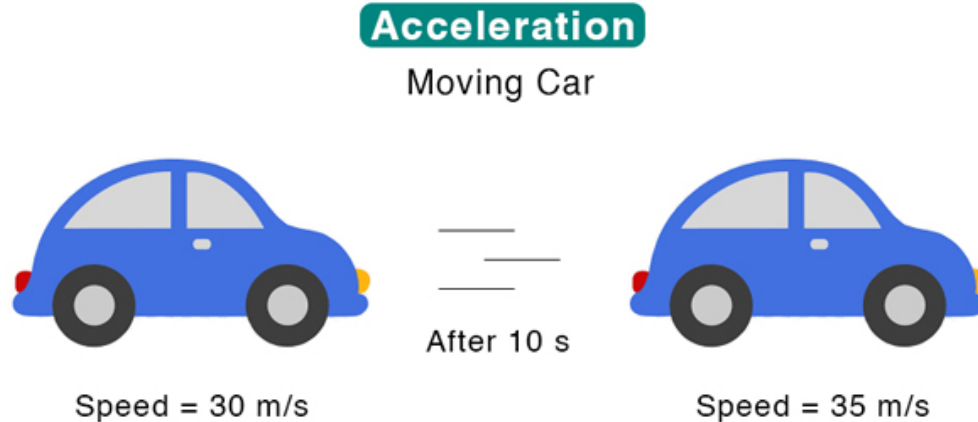
GENERAL DEFINITIONS

GENERAL DEFINITIONS

WHAT IS ACCELERATION ??

GENERAL DEFINITIONS

ACCELERATION : The rate of change of velocity, like speeding up during takeoff, measured in meters per second squared.



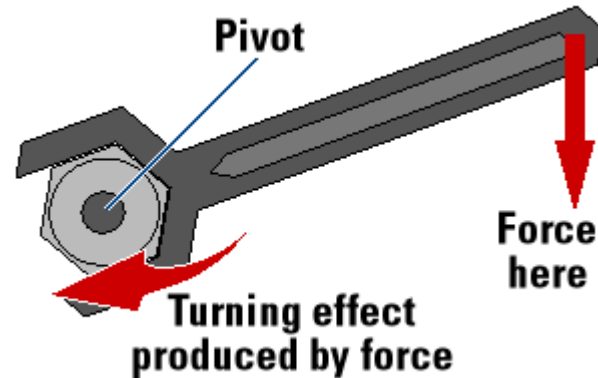
GENERAL DEFINITIONS

GENERAL DEFINITIONS

WHAT IS MOMENT ??

GENERAL DEFINITIONS

MOMENT : A turning effect caused by a force applied at a distance.



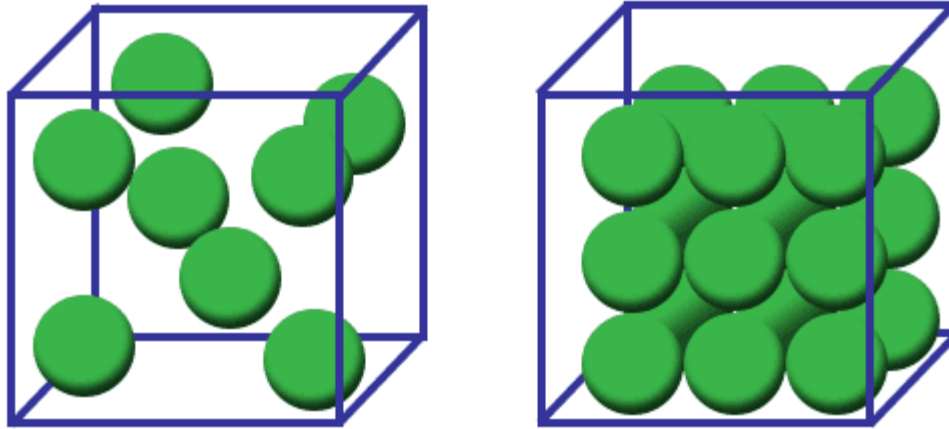
GENERAL DEFINITIONS

GENERAL DEFINITIONS

WHAT IS AIR DENSITY ??

GENERAL DEFINITIONS

AIR DENSITY : The mass of air per unit volume (kg/m^3).



GENERAL DEFINITIONS

GENERAL DEFINITIONS

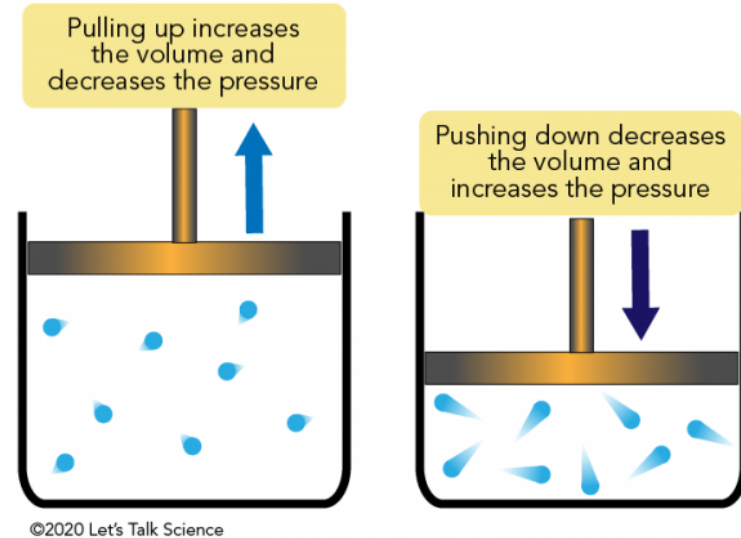
WHAT IS AIR PRESSURE ??

GENERAL DEFINITIONS

AIR PRESSURE : The push that air exerts on a surface, caused by air molecules bumping into it.

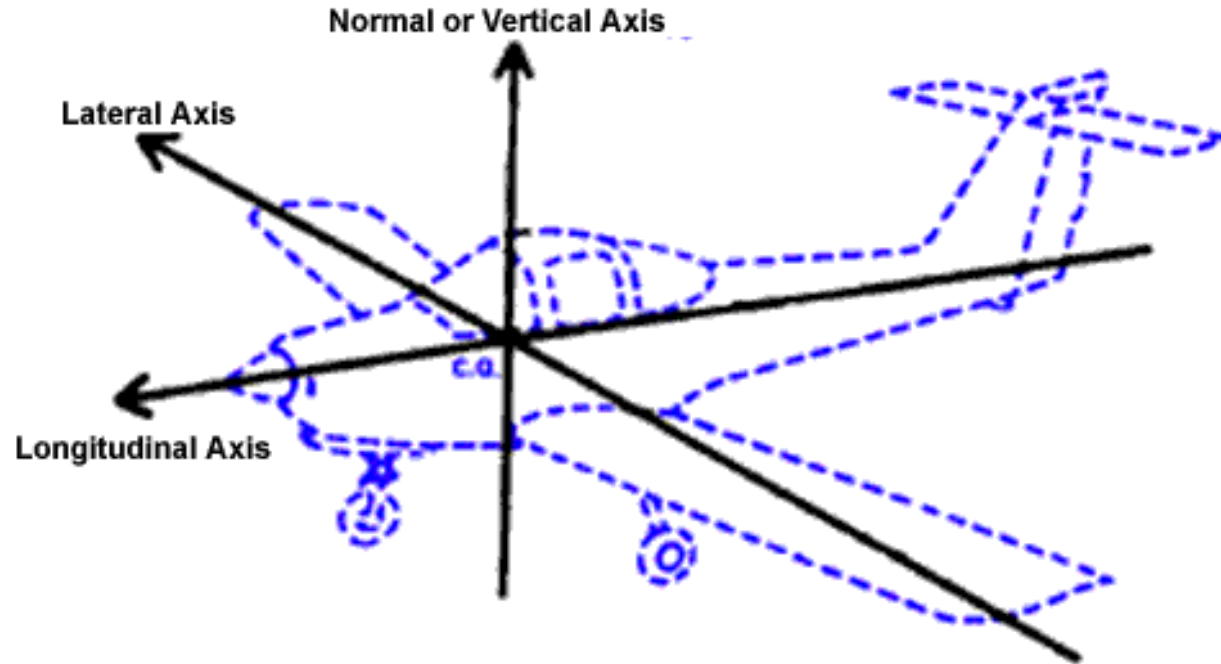
Imagine air as a bunch of tiny particles zooming around, constantly hitting everything—like a crowd gently bumping into you from all sides.

Air pressure is how much those bumps push on something, like the wings of an airplane. It's measured as force over an area (newtons per square meter).



AIRCRAFT AXES

GENERAL DEFINITIONS



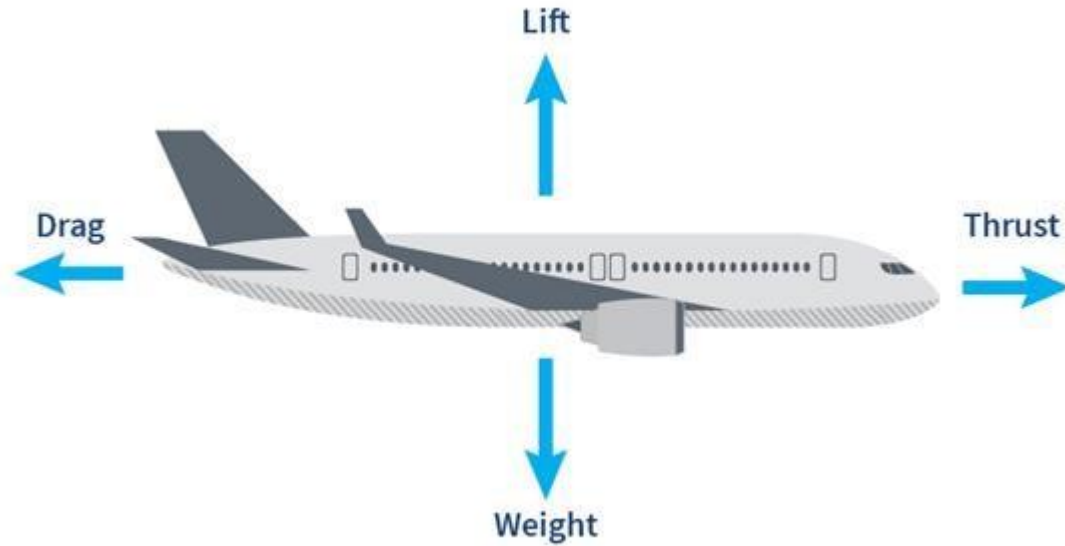
The Axes of an Airplane

AIRCRAFT FORCES

GENERAL DEFINITIONS

**WHAT ARE THE FORCES THAT AFFECT
AN AIRCRAFT ??**

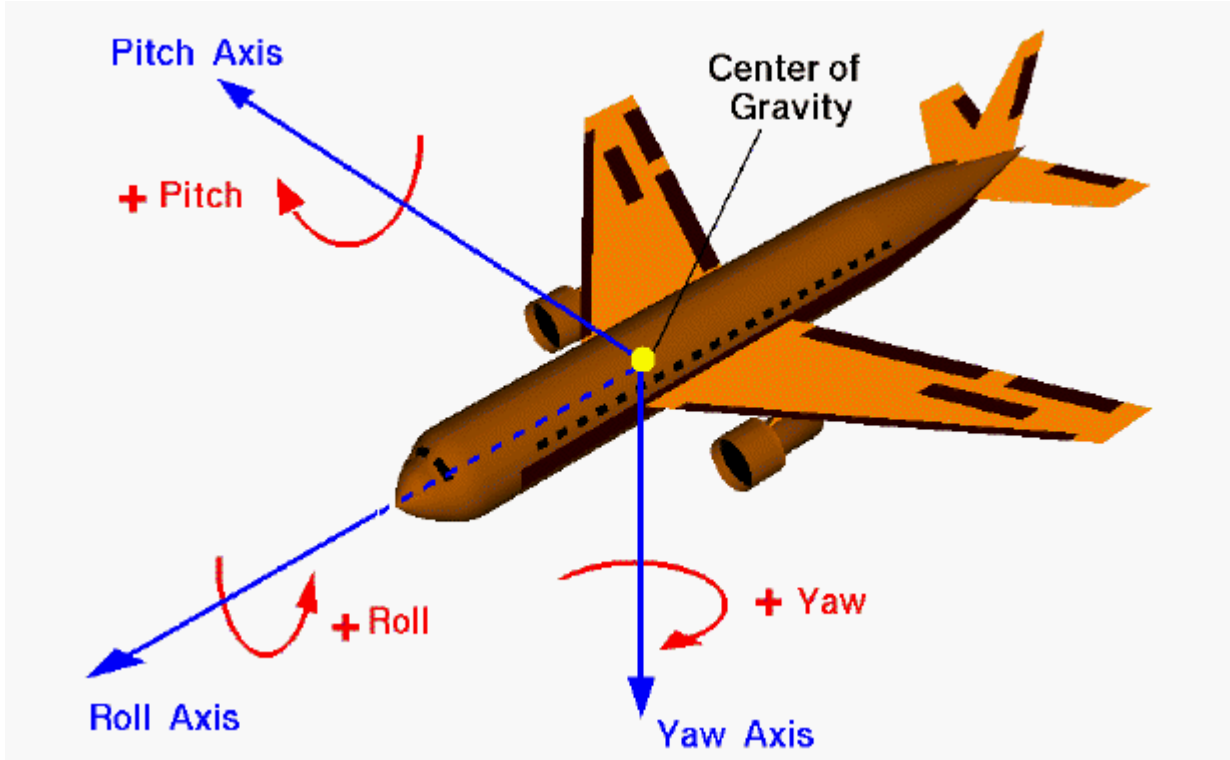
AIRCRAFT FORCES



GENERAL DEFINITIONS

AIRCRAFT MOVEMENTS

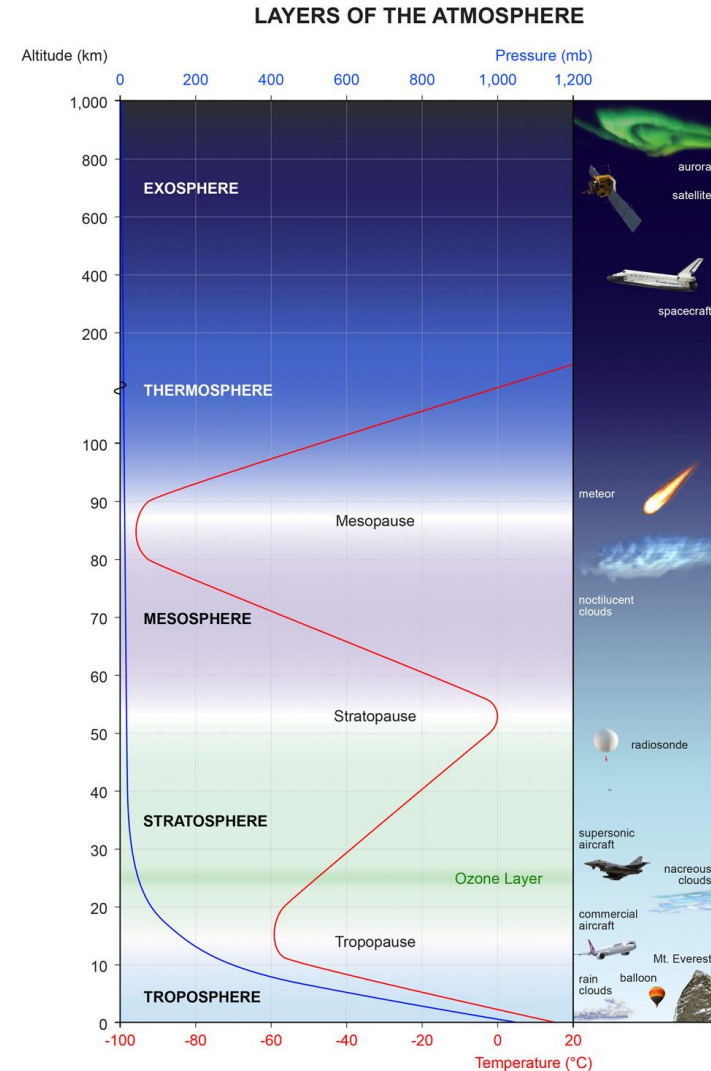
GENERAL DEFINITIONS



ATMOSPHERE

Why Study the Atmosphere?

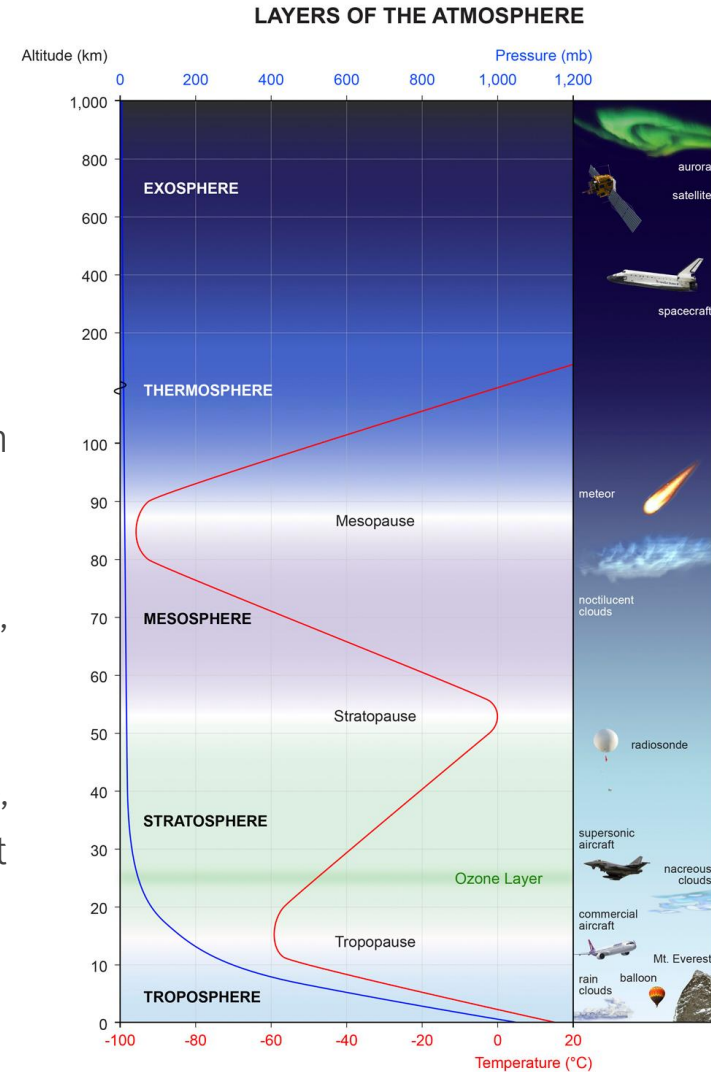
- Air is the working medium for flight—changes in **temperature, pressure, and density** affect lift, drag, and engine performance.



TROPOSPHERE

The Troposphere: Where Aviation Lives

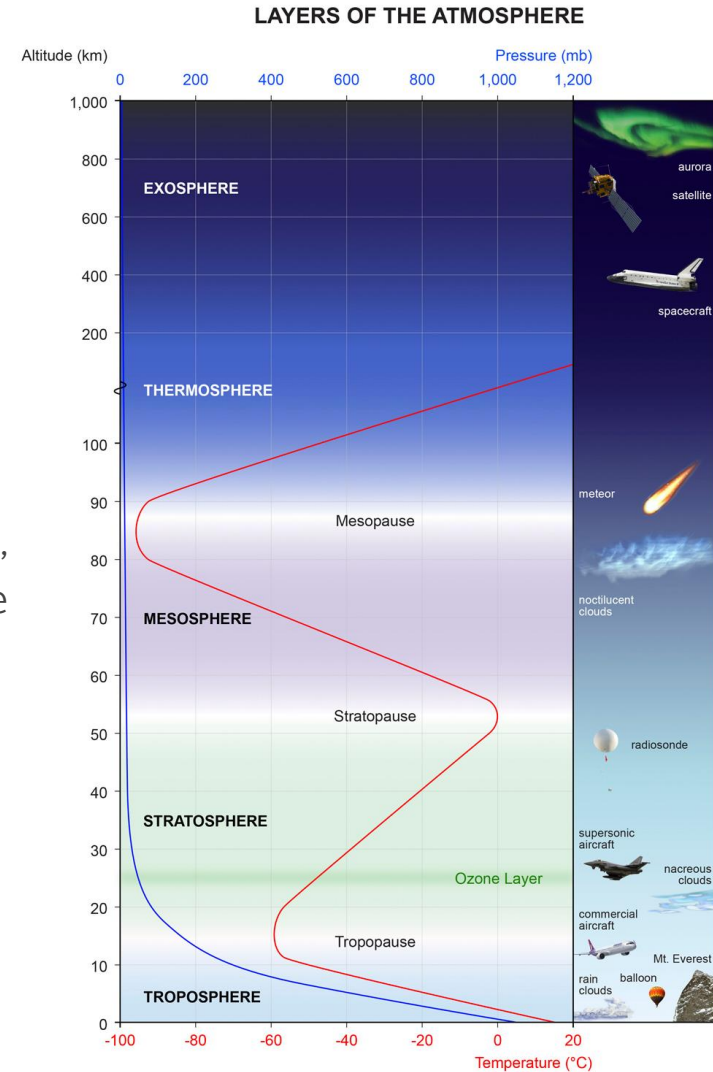
- The troposphere extends from sea level up to about 11-12 km (around 36,000 feet)
- This is the layer where commercial jets cruise, weather forms, and most of our flying occurs.
- Air properties here change dramatically with altitude, impacting aircraft in real ways—think reduced engine thrust at high altitudes or the need for pressurized cabins.



TROPOSPHERE

Temperature Changes

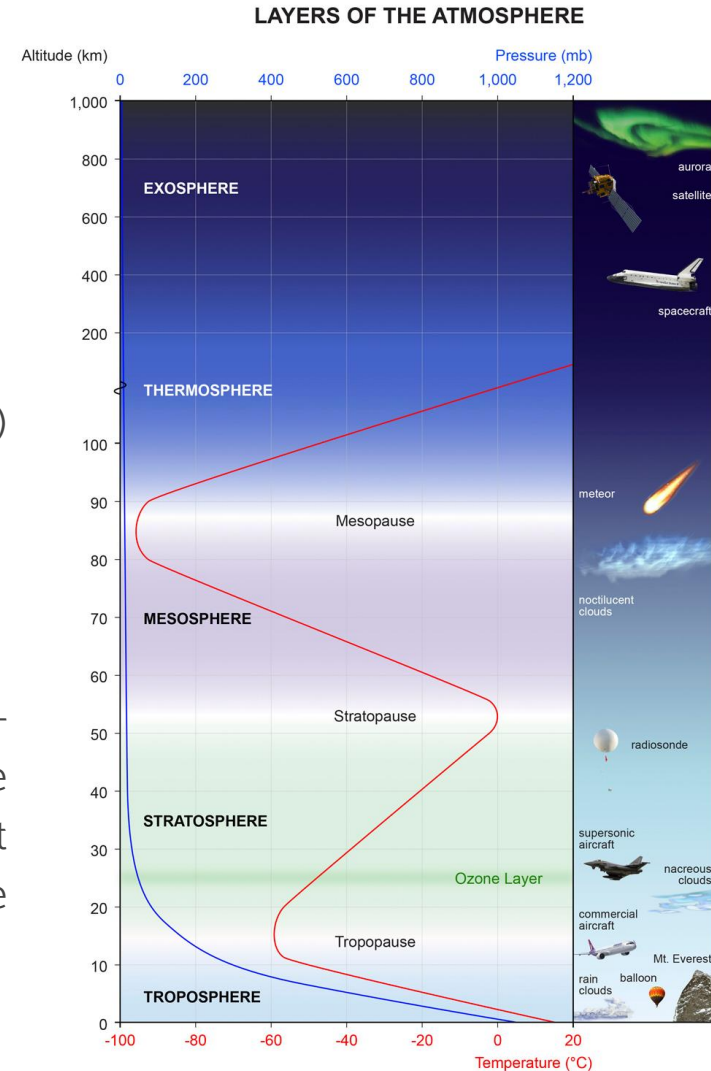
- Temperature decreases steadily as altitude increases, dropping from about 15°C at sea level to around -56°C at the top



TROPOSPHERE

Pressure Changes

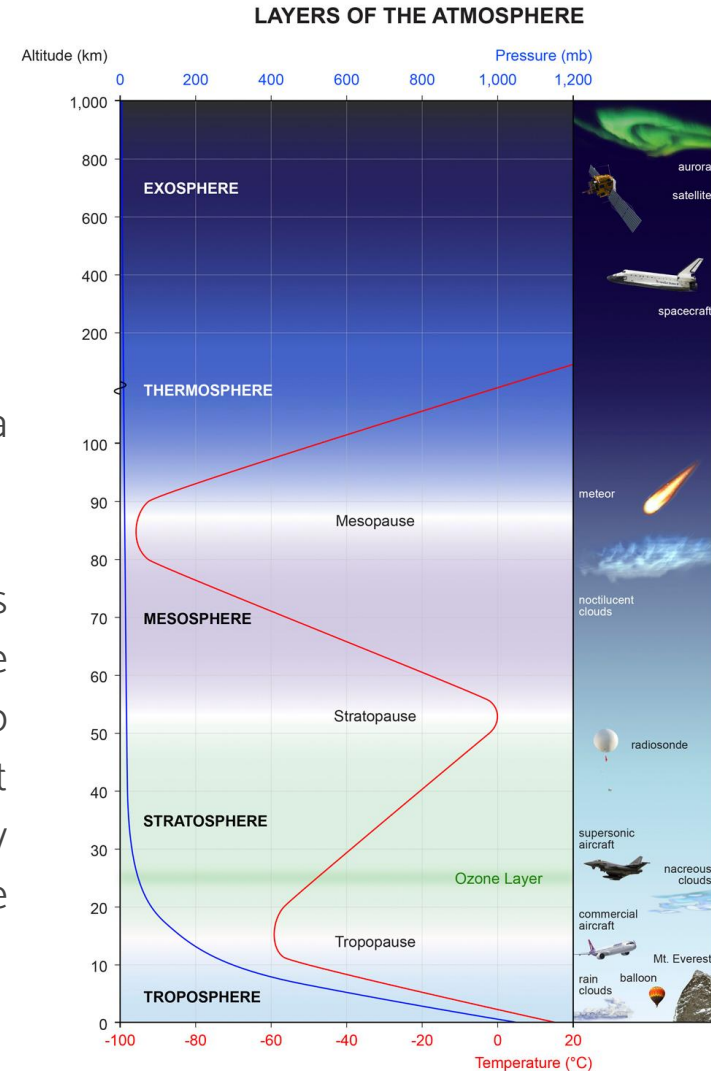
- Decreases rapidly with altitude, from about 1013 millibars (mb) at sea level to roughly 200 mb at the top
- it's not linear but exponential, halving about every 5.5 km.
- Lower pressure means less oxygen for engines and humans—jets use compressors to maintain thrust, and cabins are pressurized to simulate lower altitudes (around 8,000 feet inside). Altimeters rely on this pressure drop to measure height.



TROPOSPHERE

Pressure Changes

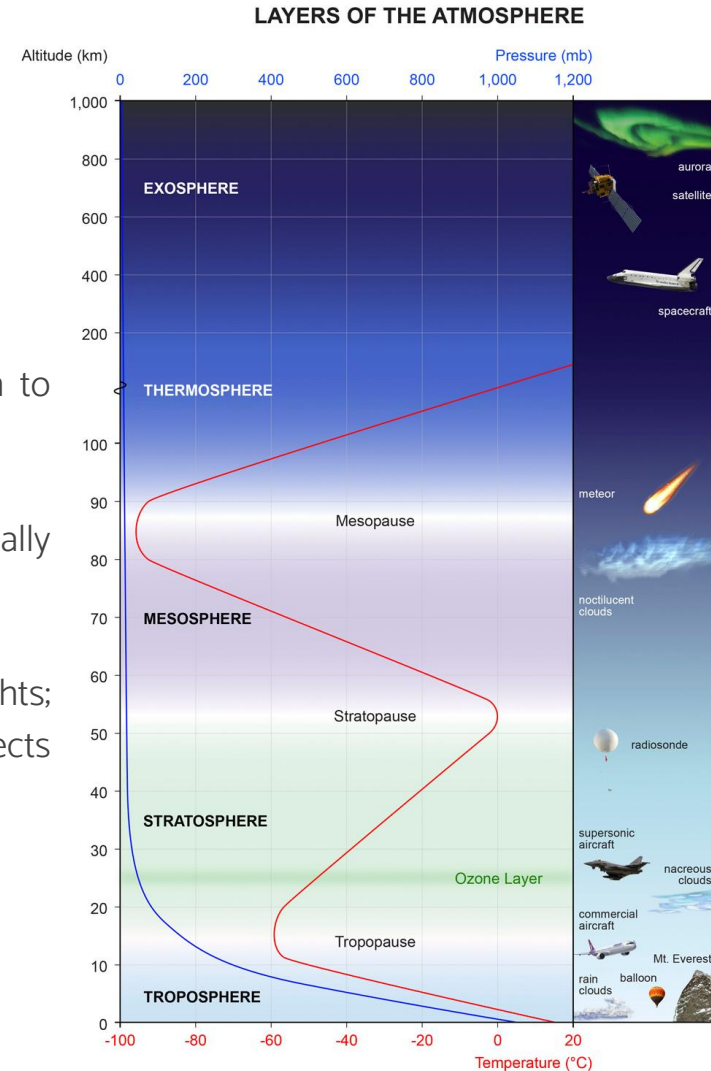
- Density decreases with altitude, from about 1.2 kg/m^3 at sea level to around 0.3 kg/m^3 at the top
- Lift from wings depends on air density—thinner air means planes need higher speeds or larger wings to generate the same lift. Engines produce less thrust in low-density air due to fewer oxygen molecules for combustion. This is why airports at high elevations (like Denver) have longer runways, and why jets climb to cruising altitudes where drag is lower despite thinner air.



STRATOSPHERE

Stratosphere (About 12-50 km)

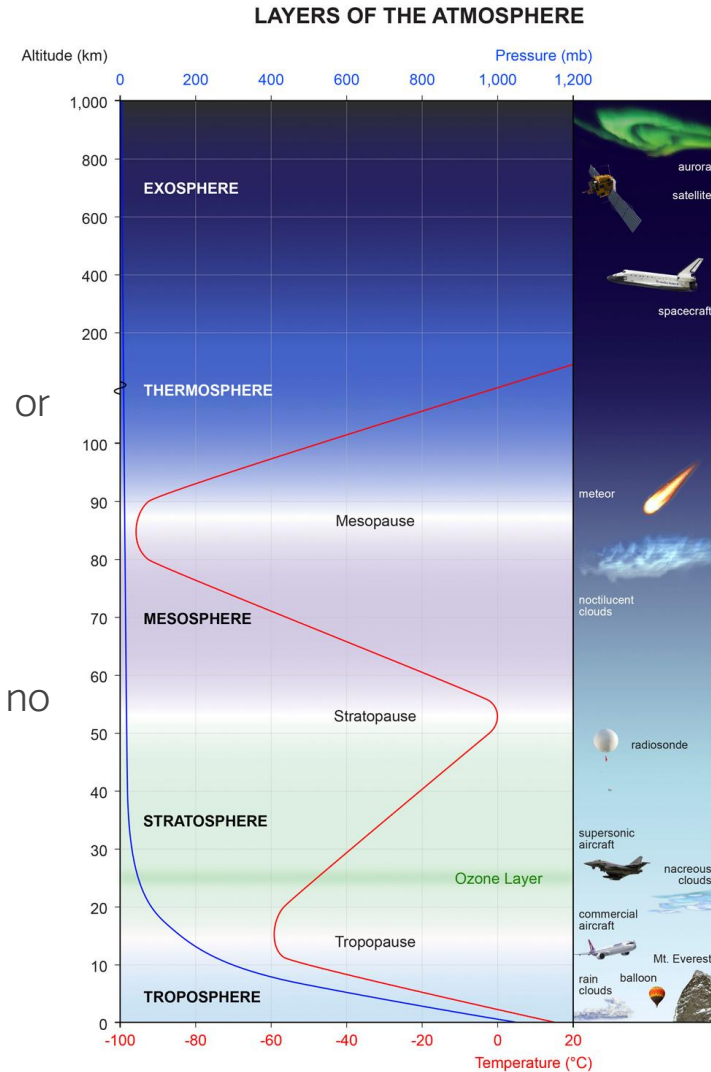
- **Temperature:** Increases with altitude, from -56°C at the bottom to around 0°C at the top.
- **Pressure and Density:** Both decrease further, but more gradually than in the troposphere.
- **Aviation note:** Concorde flew here for smoother, faster flights; modern jets sometimes clip the lower edge. The ozone layer protects against UV, but thin air limits conventional flight.



MESOSPHERE

Mesosphere (About 50-85 km)

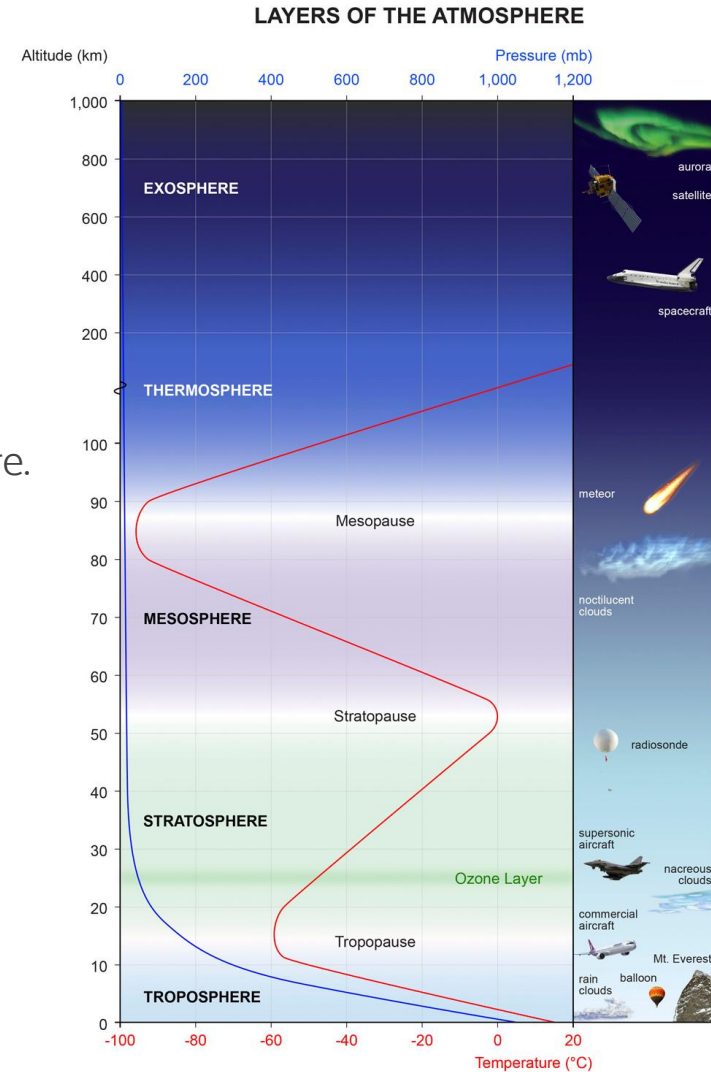
- **Temperature:** Decreases sharply again, down to -90°C or colder
- **Pressure and Density:** Extremely low—air is very sparse.
- **Aviation note:** Meteors burn up here due to friction; no aircraft operate this high.



THERMOSPHERE

Thermosphere (About 85-600 km)

- **Temperature:** Increases dramatically, up to $1,500^{\circ}\text{C}$ or more.
- **Pressure and Density:** Near vacuum levels.
- **Aviation note:** Satellites and space shuttles orbit here; auroras form from solar particle interactions.



THERMOSPHERE

Exosphere (Above 600 km)

- **Temperature:** Variable and high, but meaningless in traditional sense.
- **Pressure and Density:** Essentially zero.
- **Aviation note:** This is outer space—relevant for spacecraft, not planes.

